MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION October 2026
Fifth SEMESTER
Discipline Specific Core Course (DSC) – MG5DSCCND301 – PAEDIATRIC NUTRITION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

PART A

Objective
Answer all questions
Each question carries 1 mark

1. a) Tone
2. d) Poor oral-motor coordination, swallowing dysfunction, and GER
3. a) Growth hormone–IGF-1 axis
4. b) Heavy proteinuria
5. c) To restore and maintain a healthy gut microbial population
6. a) To improve outcomes through early recognition and management
7. b) Blood and urine
8. c) Type 2 diabetes mellitus
9. b) 4 g fat: 1 g protein + carbohydrate
10. a) Metabolic acidosis

PART B

Short Answer
Answer any 5 questions
Each question carries 2 marks

11. Describe the clinical manifestation of iodine deficiency ***(1 mark each)***

- Goitre: Enlargement of the thyroid gland located at the front of the neck.
- Other manifestations: Cretinism (physical and mental retardation) and deaf-mutism (hearing and speech impairment).

12. State the role of a small intestinal biopsy in diagnosing celiac disease. ***(1 mark each)***

- Used to confirm celiac disease by collecting a tissue sample during upper gastrointestinal endoscopy.
- Microscopic examination identifies intestinal villous damage or flattening, which leads to nutrient malabsorption.

13. List any four common food allergens. ***(Any 4, 1/2 mark each)***

- Cow's milk, egg, peanut, tree nuts, soy, wheat, fish, shellfish and sesame.

14. Explain how defective CFTR function results in thick, sticky secretions. ***(1 mark each)***

- Defective CFTR protein function impairs chloride ion transport across epithelial cells, disrupting the normal movement of sodium and water.
- Reduced water movement into secretions makes them dehydrated, thick, and sticky, causing obstruction of the airways and ducts of various organs.

15. Define epilepsy. ***(1 mark each)***

- Epilepsy is a chronic neurological disorder characterized by a persistent tendency to experience recurrent, unprovoked seizures.
- These seizures occur due to abnormal, excessive, or synchronous electrical activity in the brain.

16. Describe the procedure for recording and plotting measurements on a WHO Growth Chart. ***(1/2 mark for each point)***

- Select the appropriate WHO growth chart based on the child's sex, age, and indicator being assessed, such as weight-for-age or height-for-age.
- Accurately measure the child's weight using a calibrated weighing scale, length for children under 2 years using an infantometer, or height for children aged 2 years and above using a stadiometer.
- Plot the measurement by marking the child's age on the X-axis and the corresponding anthropometric measurement on the Y-axis.
- Record measurements at each follow-up visit, connect consecutive points to form a growth curve, and interpret the pattern using WHO Z-score curves or percentile lines.

17. Explain gastroesophageal reflux in preterm infants and mention any two management measures.

Gastroesophageal reflux ***(1/2 mark each)***

- Gastroesophageal reflux occurs when stomach contents flow back into the esophagus.
- It commonly results from immature gastrointestinal function and poor control of the lower esophageal sphincter.

Management ***(Any 2 points, 1/2 mark each)***

- Ensure proper positioning during and after feeding.
- Avoid overfeeding and adjust feeding volumes appropriately.
- Provide smaller, more frequent feeds.
- Severe cases require medical evaluation and appropriate treatment.

Part C

Short Essay
Answer any 6 questions

Each question carries 5 marks

18. Define nutritional assessment in peadiatrics and explain any two parameters of anthropometry.

- Nutritional assessment in paediatrics is the systematic evaluation of a child's nutritional status and growth to identify, prevent, and manage malnutrition. **(*1 mark)***

Parameters of Anthropometric Assessment in Paediatrics *(Any 2 parameters- 2 mark each)*

1. **Height/Length (*1/2 mark for each point)***

- Length is measured in children below 2 years using an infantometer, with the child lying down (known as recumbent or crown-heel length).
- The child's head is placed against the fixed headpiece, the body is kept straight, and the knees and ankles are gently pressed flat against the board.
- Standing height (stature) is measured in children aged 2 years and above using a stadiometer or wall-mounted measuring board, with bare feet and the child standing erect.
- The head is positioned horizontally, and the measurement is recorded to the nearest 0.1 cm.

2. **Weight (*(1/2 mark for each point)***

- Body weight is one of the simplest and most widely used anthropometric measurements for assessing nutritional status.
- Weight is measured using suitable weighing scales, such as Salter scales, beam-balance scales, or digital scales.
- The child should be weighed with minimal clothing, without shoes, and without holding any support.
- For infants who cannot be weighed separately, the caregiver can be weighed while holding the infant, and the caregiver's weight is subtracted to obtain the infant's weight.

3. **Mid-Upper Arm Circumference (MUAC) (*1/2 mark for each point)***

- MUAC indicates muscle development and helps identify wasting and protein-energy malnutrition.
- It is measured on the left upper arm using a flexible, non-stretchable tape.
- With the forearm bent at a right angle, mark the midpoint between the acromion process and the tip of the ulna.
- Place the tape around the midpoint without compressing the tissues and record the measurement to the nearest 0.1 cm.

4. **Head Circumference (*1/2 mark for each point)***

- Head circumference indicates head and brain growth, especially in children up to 3 years.
- It is measured using a narrow, non-stretchable measuring tape.

- Pass the tape above the eyebrows, around the sides of the head, and over the most prominent part of the occiput.
- Adjust the tape to obtain the maximum circumference, pressing the hair against the skull, and record the measurement accurately.

5. Skinfold Thickness (*1/2 mark for each point)*

- Skinfold thickness is used to estimate body fat and assess the child's fat reserves.
- Common measurement sites include the triceps, subscapular, and suprailiac regions.
- The skin and underlying fat are gently pinched between the thumb and forefinger without including muscle, and the fold is measured using a skinfold caliper.
- Three measurements are recommended, and their average is calculated for greater accuracy.

19. Explain the types of cyanotic congenital heart disease with suitable examples.

Cyanotic congenital heart disease (1 mark)

- Cyanotic congenital heart disease is a group of heart defects present at birth that cause low oxygen levels in the blood.
- It results in bluish discoloration of the skin and may require early medical treatment or surgery.

Types

1. **Left Heart Obstructive Lesions *(1 mark)***

- These defects reduce blood flow from the left side of the heart to the rest of the body (systemic circulation). Examples include:
  - Hypoplastic left heart syndrome: Underdevelopment of the left side of the heart, reducing blood flow to the body.
  - Interrupted aortic arch: Incomplete development of the aorta, obstructing blood flow to the body.

2. Right Heart Obstructive Lesions *(2 mark)*

- These defects reduce blood flow from the heart to the lungs (pulmonary circulation), affecting oxygenation of blood. Examples include:
  - Tetralogy of Fallot: Four heart abnormalities that reduce blood flow to the lungs and cause cyanosis.
  - Ebstein anomaly: Abnormal tricuspid valve development affecting blood flow through the right side of the heart.
  - Pulmonary atresia: Failure of the pulmonary valve to develop properly, blocking blood flow to the lungs.
  - Tricuspid atresia: Absence or abnormal development of the tricuspid valve, restricting blood flow through the right heart.

3. **Mixing Lesions *(1 mark)***

- These defects allow oxygen-rich and oxygen-poor blood to mix, reducing the oxygen supply delivered to the body. Examples include:

- Transposition of the great arteries: The aorta and pulmonary artery arise from the wrong ventricles, causing abnormal circulation.
- Truncus arteriosus: A single major artery arises from the heart instead of separate aorta and pulmonary artery.

20. Describe the different types of dental caries seen in children. ***(Any 5 points, 1 mark each)***

Types of Dental Caries in Children

- Early Childhood Caries (ECC): Dental decay in primary teeth, commonly associated with frequent sugar exposure.
- Nursing Bottle Caries: Decay caused by prolonged exposure to milk, formula, juice, or other sugary drinks, especially during sleep with a bottle.
- Pit and Fissure Caries: Decay in pits and grooves, particularly on the chewing surfaces of molars.
- Smooth Surface Caries: Decay on smooth tooth surfaces due to plaque and prolonged sugar exposure.
- Root Caries: Decay affecting exposed tooth-root surfaces; uncommon in young children.
- According to stage of development:

- Initial/Early Caries: Enamel demineralization appearing as white spots.
- Established Caries: Enamel destruction with cavity formation.
- Advanced Caries: Decay spreads into dentin and pulp, potentially causing pain and sensitivity.

21. Describe the diagnosis of diarrhoea in children.

1. **History** ***(1 mark)***

- Assess duration, frequency, stool consistency, blood or mucus, vomiting, fever, and abdominal pain.
- Record fluid intake, urine output, dietary history, antibiotic use, travel, contaminated food/water exposure, weight loss, and previous episodes.

2. Physical Examination ***(1 mark)***

- Assess general appearance, consciousness, and hydration status.
- Check skin turgor, sunken eyes, dry mouth, thirst, pulse, circulation, and urine output.
- Assess weight, growth pattern, and nutritional status.

3. Laboratory Investigations ***(2 mark)***

- Blood tests to assess dehydration, electrolyte imbalance, anaemia, inflammation, and nutritional status.
- Stool examination to identify pathogens and parasites, especially in bloody or persistent diarrhoea.

4. Other Investigations in Chronic Diarrhoea ***(1 mark, any 2 points ½ mark each)***

- Endoscopy or colonoscopy with biopsy to identify intestinal abnormalities.

- Lactose hydrogen breath test to detect lactose malabsorption.
- Imaging when indicated to investigate underlying gastrointestinal conditions.

22. Describe the different assistive feeding devices used for children with developmental disabilities and explain how they promote safe and independent feeding. ***(1 mark for each point)***

1. Cups

- Cut-out cups prevent backward head tilting, while small-diameter and handled cups improve grip and control.
- Spouted cups may help during bottle transition, while spout-free lids reduce spills, control liquid flow, and encourage lip closure.

2. Straws

- Improve sucking skills, lip strength, and oral control while promoting proper drinking posture.
- Colourful or fun straws encourage children to practise drinking.

3. Spoons

- Shallow spoons ease food removal, while rubber-coated or plastic spoons are gentle on sensitive mouths.
- Built-up or bent handles improve grip and control; spoon cuffs or holders secure the spoon for self-feeding.

4. Plates and Bowls

- Plate guards and scoop dishes help children collect food easily.
- Suction-base dishes and non-slip mats prevent plates and bowls from sliding.

5. Seating and Positioning

- Supportive chairs, footrests, trays, and cushions provide head, neck, and trunk stability and improve hand-to-mouth coordination.
- Proper positioning improves feeding efficiency, reduces fatigue, and lowers choking and aspiration risks.

23. Examine the short term consequences of undernutrition. ***(1 mark for each point)***

1. Growth Impairment

- Inadequate energy, protein and essential nutrients reduce IGF-1 production and slow normal growth, resulting in poor weight gain and reduced growth velocity.
- If undernutrition persists, it may lead to growth faltering, stunting and failure to thrive.

2. Weakened Immune System

- Undernutrition impairs innate and adaptive immunity, increasing susceptibility to infections such as diarrhoea and respiratory infections.
- Infections reduce appetite and increase nutrient requirements, further worsening undernutrition and creating an infection–undernutrition cycle.

3. Fatigue and Weakness

- Inadequate energy intake causes the body to use stored fat and, during prolonged deficiency, muscle tissue for energy.

- Loss of energy reserves and muscle mass leads to fatigue, weakness, reduced physical activity and poor physical capacity.

4. **Weight Loss and Wasting**

- Acute or severe undernutrition causes rapid weight loss and wasting, characterised by low weight-for-height.
- Wasting indicates recent nutritional deprivation or illness and is associated with impaired immunity, increased complications and mortality risk.

5. **Increased Susceptibility to Infection and Mortality**

- Undernourished children have a higher risk of frequent infections, particularly diarrhoeal and respiratory infections, due to impaired immunity and inadequate body reserves.
- Severe undernutrition, especially wasting, increases the risk of serious complications and death.

24. Describe the laboratory investigations used in the diagnosis of T1DM. ***(1 mark for each point)***

1. **Blood Glucose Tests**

- Fasting plasma glucose (FPG) ≥126 mg/dL after at least 8 hours of fasting.
- Random plasma glucose ≥200 mg/dL in the presence of classic symptoms, or 2-hour plasma glucose ≥200 mg/dL during a 75-g oral glucose tolerance test (OGTT).
- HbA1c ≥6.5% indicates diabetes.

2. **Autoantibody Testing**

- Autoantibodies help confirm autoimmune T1DM.
- Common antibodies include GAD65, IA-2, insulin autoantibody (IAA), ZnT8 and islet cell antibody (ICA).

3. **C-Peptide Measurement**

- C-peptide indicates the body's endogenous insulin production.
- Low or absent C-peptide suggests reduced insulin secretion due to pancreatic beta-cell destruction.

4. **Ketone Testing**

- Urine or blood ketone tests help identify ketone accumulation.
- These tests are important for detecting diabetic ketoacidosis (DKA), a serious complication of T1DM.

5. **Additional Investigations**

- Serum electrolytes are assessed to identify electrolyte imbalances.
- Renal function tests help evaluate kidney function, particularly during acute illness or DKA.

25. Define lactose intolerance and explain any two methods used to diagnose lactose intolerance.

Lactose intolerance ***(1 mark, ½ mark for each point)***

- The inability to digest lactose, the natural sugar present in milk and dairy products, due to insufficient production of the enzyme lactase in the small intestine.
- Undigested lactose passes into the large intestine, where bacteria ferment it, causing symptoms such as bloating, gas and diarrhoea.

Diagnosis ***(Any 2 points, 2 marks for each)***

1. **Hydrogen Breath Test** ***(1/2 mark for each point)***

- It is the most commonly used and reliable test for diagnosing lactose intolerance.
- The individual drinks a lactose-containing solution, and hydrogen and methane levels in the exhaled breath are measured at regular intervals.
- Undigested lactose passes into the large intestine, where intestinal bacteria ferment it and produce hydrogen and methane gases.
- Elevated hydrogen or methane levels in the breath indicate lactose malabsorption and support the diagnosis of lactose intolerance.

2. **Blood Glucose Test** ***(1/2 mark for each point)***

- This test may be used when the hydrogen breath test does not provide clear results.
- Blood glucose levels are measured before and after the individual consumes a lactose-containing solution.
- Normally, lactase breaks down lactose into glucose and galactose, resulting in an increase in blood glucose levels.
- Little or no rise in blood glucose indicates inadequate lactose digestion and absorption, supporting the diagnosis of lactose intolerance.

3. **Stool Acidity Test** ***(1/2 mark for each point)***

- This test is mainly used in infants and young children who may not be able to perform the hydrogen breath test.
- A stool sample is collected and examined after the child consumes lactose.
- Undigested lactose is fermented by intestinal bacteria, producing lactic acid and other organic acids.
- A low stool pH indicates increased acidity due to organic acid production and supports the diagnosis of lactose malabsorption.

Part D

Essay
Answer any 2 questions
Each question carries 10 marks

26. Evaluate the role of specialised diets used in the management of food allergy, with reference to elimination diets and other dietary management approaches, nutritional monitoring and regular follow-up.

Role of Specialised Diets in Food Allergy (*6 Marks, 2 mark for each heading*)

1. **Individualized Elimination Diet** ***(2 marks)***

- Elimination diets are the cornerstone of dietary management of confirmed food allergies and should be individualized according to the type and severity of allergy, number of allergens, age, nutritional status and affordability of alternatives.
- A single-food elimination diet removes only the confirmed offending food while retaining other tolerated foods, helping maintain dietary variety and reduce nutritional risks.
- A multiple-food elimination diet removes two or more confirmed allergens, requiring monitoring for inadequate energy, protein and micronutrient intake, feeding difficulties and growth faltering.
- Extensive or few-food elimination diets are reserved for selected diagnostic or severe non-IgE-mediated conditions under specialist supervision because of their nutritional and psychosocial risks.

2. **Specialized Diets for Eosinophilic Esophagitis (EoE)** ***(2 marks)***

- One-food elimination diet (1FED): Usually eliminates cow's milk, allowing greater dietary variety and better adherence.
- Two-food elimination diet (2FED): Commonly eliminates cow's milk and wheat.
- Four-food elimination diet (4FED): Commonly eliminates dairy, soy, egg and wheat; it may be followed for approximately 8–12 weeks before reassessment and supervised food reintroduction.
- More extensive elimination diets: Six-food or broader diets may be used in selected cases, but increase nutritional risks and make adherence difficult.
- Elemental diet: Uses an amino-acid-based formula as the primary nutritional source and is reserved for selected cases under specialist medical and dietetic supervision.

3. **Dietary Management of Food Protein-Induced Enterocolitis Syndrome (FPIES)** ***(2 marks)***

- Identify and avoid confirmed trigger foods that cause gastrointestinal symptoms, particularly repetitive vomiting and diarrhoea.
- Provide safe alternative foods to maintain adequate energy and protein intake.
- Maintain dietary diversity and prevent unnecessary restriction of multiple foods.
- Monitor nutritional status and growth regularly to prevent nutritional deficiencies and growth faltering.

Other dietary management approaches (4 marks, ½ mark for each point)

1. **Complementary Feeding**

- Introduce age-appropriate fruits, vegetables, iron-rich foods, cereals and other tolerated foods with suitable texture progression, avoiding unnecessary food restrictions.

2. Food Label Reading

- Educate parents and children to read ingredient lists, identify hidden allergens and alternative names, and understand precautionary statements such as "may contain."

3. Cross-Contact Prevention

- Prevent accidental allergen transfer by using clean utensils, surfaces and cooking equipment, and educating children, caregivers and teachers about allergen avoidance.

4. Nutritional Substitution

- Follow the principle of "remove and replace" by providing safe alternatives that maintain adequate energy, protein, calcium, vitamin D and other essential nutrients.

5. Growth and Nutritional Monitoring

- Regularly monitor weight, height/length, BMI-for-age, dietary diversity and nutrient adequacy, with supplementation or biochemical assessment when clinically indicated.

6. Feeding Difficulties

- Address food refusal, fear of eating, limited food variety and texture difficulties through positive feeding practices and appropriate feeding or psychological support.

7. School and Social Activities

- Educate teachers, caregivers and food-service personnel about allergen avoidance and emergency plans, enabling safe participation in school and social activities.

8. Education and Follow-up

- Educate families about allergen avoidance, symptom recognition, cross-contact prevention and epinephrine use, with regular follow-up to reassess food tolerance when appropriate.

27. Explain the mineral and vitamin requirements in preterm and low birth weight infants.

- Nutrient requirements in preterm and LBW babies are higher because of rapid growth, immature organ function and limited nutrient stores.
- Nutrients are provided through parenteral nutrition (PN) or enteral nutrition, depending on the infant's clinical condition and feeding tolerance.

Mineral and Vitamin Requirements (Any 10 nutrients, 1 mark for each point)

1. Sodium (Na^+)

- Maintains extracellular fluid balance, nerve conduction, normal growth and development.
- Parenteral nutrition (PN): Usually provided after the first few days of life to replace losses; approximately 2–3 mEq/kg/day, individualized according to serum sodium, hydration and renal function.

- Enteral nutrition: Approximately 2–3 mEq/kg/day; very low birth weight (VLBW) infants with increased renal losses or hyponatremia may require 4–8 mEq/kg/day or more under medical supervision.

2. Potassium (K^+)

- Maintains intracellular fluid balance, neuromuscular function and normal cardiac activity.
- Parenteral nutrition (PN): Approximately 2–3 mEq/kg/day; introduced only after adequate urine output and renal function are established to prevent hyperkalemia.
- Enteral nutrition: Approximately 2–3 mEq/kg/day, supplied through human milk or preterm formula.

3. Chloride (Cl^-)

- Maintains acid–base balance, osmotic pressure and fluid regulation.
- Parenteral nutrition (PN): Added with sodium and potassium after the initial days of life to maintain electrolyte and acid–base balance; requirements are individualized.
- Enteral nutrition: Usually supplied along with sodium and potassium through human milk or fortified human milk and preterm formula.

4. Calcium

- Essential for bone mineralization, skeletal growth, muscle contraction and nerve function.
- Parenteral nutrition (PN): Approximately 500–600 mg/L, administered with phosphorus; serum calcium and alkaline phosphatase levels are monitored.
- Enteral nutrition: Approximately 165–180 mg/100 kcal; fortified human milk or preterm formula helps prevent osteopenia of prematurity.

5. Phosphorus

- Required for bone mineralization, ATP production, energy metabolism and normal cellular function.
- Parenteral nutrition (PN): Approximately 400–500 mg/L, administered with calcium; serum phosphorus and alkaline phosphatase levels are monitored.
- Enteral nutrition: Approximately 82–100 mg/100 kcal; adequate intake supports bone growth and prevents metabolic bone disease.

6. Iron

- Essential for haemoglobin synthesis, red blood cell formation, oxygen transport and neurodevelopment.
- Parenteral nutrition (PN): Not routinely provided; when indicated, approximately 0.1–0.2 mg/kg/day may be used according to clinical requirements.
- Enteral nutrition: Approximately 2–4 mg/kg/day after two weeks of age; infants receiving erythropoietin may require higher amounts, and breastfed preterm infants generally require supplementation.

7. Zinc

- Supports growth, immune function, wound healing and enzyme activity.
- Parenteral nutrition (PN): Approximately 400 µg/kg/day, particularly important during prolonged PN.

- Enteral nutrition: Approximately 1–3 mg/kg/day, usually supplied through fortified human milk or preterm formula.

8. Vitamin A

- Essential for vision, epithelial integrity, immune function and lung development.
- Parenteral nutrition (PN): Provided through paediatric intravenous multivitamin preparations. Recommendation: 700-1500 IU/Kg/day.
- Enteral nutrition: Approximately 1300–3300 IU/kg/day, generally supplied through fortified human milk or preterm formula.

9. Vitamin D

- Promotes calcium absorption, bone mineralization and prevention of metabolic bone disease.
- Parenteral nutrition (PN): Provided through paediatric intravenous multivitamin preparations.
- Enteral nutrition: Approximately 400-700 IU/day according to the supplied material; requirements should be individualized based on current neonatal guidance and total intake.

10. Vitamin E

- Acts as an antioxidant, protecting cells from oxidative damage and helping prevent haemolytic anaemia.
- Parenteral nutrition (PN): Included in paediatric intravenous multivitamin preparations.
- Enteral nutrition: Approximately 6–12 IU/kg/day; pharmacological doses are avoided because of possible adverse effects.

11. Vitamin K

- Essential for blood coagulation and prevention of haemorrhagic disease of the newborn.
- Parenteral nutrition (PN): Vitamin K prophylaxis is given intramuscularly at birth, commonly 0.5–1 mg as specified in the supplied material; subsequent provision may occur through intravenous multivitamin preparations.
- Enteral nutrition: Routine prophylaxis is given at birth; subsequent intake is supplied through human milk or fortified preterm formula, with supplementation according to clinical guidance.

28. Explain the different standards used for interpreting anthropometric data in children, with reference to IAP, Gomez, and NCHS standards.

1. IAP Standards *(1 Mark)*

- The Indian Academy of Paediatrics (IAP) growth charts are used to assess the growth and nutritional status of children in India.
- Used for children aged 5-18 years.

Steps in interpretation *(1 mark)*

- Measure the child's age, sex, weight, height and BMI.
- Plot the measurements on the appropriate growth chart.
- Compare the values with percentile curves or Z-scores.

- Interpret the child's nutritional status based on the position on the chart.

Interpretation of anthropometric parameters *(2 marks)*

- Height-for-age: Below the 3rd percentile indicates short stature or stunting; between the 3rd and 97th percentiles is generally considered normal.
- Weight-for-age: Below the 3rd percentile indicates underweight; between the 3rd and 97th percentiles is generally considered normal.
- BMI-for-age (5–18 years): Used to classify thinness, normal weight, overweight and obesity. The revised IAP charts use adult-equivalent BMI cut-offs of 23 kg/m² for overweight and 27 kg/m² for obesity.
- Weight-for-height (under 5 years): Based on WHO standards; below −2 SD indicates wasting and below −3 SD indicates severe wasting, while above +2 SD indicates overweight.

2. Gomez Classification *(1 Mark)*

- Gomez classification is an early method used to assess protein-energy malnutrition based on the percentage of expected weight-for-age.
- Weight for age (%) = (Actual weight of the child/ Median weight for age) x 100

Classification *(1 Mark)*

- Normal: Above 90% of the expected weight.
- Mild malnutrition (Grade I): 76–90%.
- Moderate malnutrition (Grade II): 61–75%.
- Severe malnutrition (Grade III): Below 60%.

Limitations *(1 Mark)*

- It does not consider height and cannot distinguish wasting from stunting.
- It does not assess overweight or obesity.

3. NCHS Standards *(1 Mark)*

- The National Center for Health Statistics (NCHS) assess the growth and nutritional status of children from birth to 18 years.

Steps in interpretation *(1 mark)*

- Measure the child's age, sex, weight and height or length.
- Compare the measurements with NCHS reference values for children of the same age and sex.
- Assess nutritional status using weight-for-age, height-for-age, weight-for-height and BMI-for-age indices.

Interpretation of anthropometric indices *(1 Mark)*

- Weight-for-age: Assesses underweight.
- Height-for-age: Identifies stunting or chronic malnutrition.
- Weight-for-height: Identifies wasting or acute malnutrition and can indicate overweight.
- BMI-for-age: Helps assess thinness, overweight and obesity.

29. Explain the nutritional management of celiac disease in children, including common nutritional deficiencies, nutrient requirements, and appropriate dietary modifications.

1. **Gluten-Free Diet** ***(1 mark)***

- A strict, lifelong gluten-free diet is the only effective treatment for celiac disease.
- Completely avoid wheat, barley and rye to prevent intestinal damage and promote healing of the intestinal mucosa.
- A balanced gluten-free diet helps restore nutrient absorption and supports normal growth and development.

2. Common Nutritional Deficiencies ***(2 marks)***

- Iron: Deficiency may lead to anaemia.
- Calcium and vitamin D: Deficiency impairs bone mineralization and increases the risk of osteopenia and osteoporosis.
- Folate and vitamin B12: Deficiency affects red blood cell formation and neurological function.
- Zinc and magnesium: Deficiency may impair growth, wound healing and normal body functions.
- Fibre: Low intake may contribute to constipation and poor gut health.

3. **Nutrient Requirements** ***(2 marks)***

- Correct nutritional deficiencies and restore normal nutrient absorption.
- Promote intestinal healing, healthy growth and development, and improved bone and immune health.
- Prevent long-term complications such as anaemia and osteoporosis and improve overall quality of life.
- Energy and protein: Adequate intake supports catch-up growth, tissue repair and normal development.
- Iron: Include iron-rich foods to prevent and correct anaemia.
- Calcium and vitamin D: Ensure adequate intake for bone growth and mineralization.
- Folate, vitamin B12, zinc and magnesium: Provide adequate amounts to support blood formation, immunity and growth.
- Dietary fibre: Include fibre-rich gluten-free foods to maintain bowel health and prevent constipation.

4. **Dietary Modifications** ***(5 marks)***

- Include naturally gluten-free foods such as fruits, vegetables, legumes, nuts, seeds, meat, poultry, fish, eggs and milk if tolerated.
- Replace gluten-containing grains with rice, millet, sorghum, quinoa, buckwheat and amaranth; certified gluten-free oats may be included if uncontaminated and well tolerated.
- Limit processed gluten-free foods because they may be high in fat, sugar and salt but low in fibre and essential micronutrients.
- Read food labels carefully to identify hidden gluten sources such as wheat starch and malt, and choose appropriately labelled gluten-free products.

- Prevent cross-contamination during storage, preparation and cooking by using separate utensils, cutting boards, toasters and condiment containers when necessary.
- Check food preparation practices when eating outside the home and ensure medications, vitamins and supplements are suitable for a gluten-free diet.
- Regularly monitor weight, height, growth patterns, dietary intake and adherence to the gluten-free diet.
- Assess iron, calcium, vitamin D, folate, vitamin B12 and other nutrient levels when clinically indicated.
- Provide vitamin or mineral supplementation when deficiencies are confirmed or dietary intake is inadequate.
- Arrange regular follow-up with a paediatric gastroenterologist and registered dietitian, and provide continuous nutrition education to children and caregivers.